Vector addition

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Add displacement vectors and use a diagram or equation to explain where an object ends up.

SUCCESS CRITERIA - I can...

1. Connect tip-to-tail diagrams to signed component sums.
2. Give magnitude, units and an angle with a named reference direction.
3. Check the result against the diagram, limits and physical situation.

Core: teacher selects from 1-10. **Stretch:** 11-12 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “vector” and “component” to explain your answer. Sentence starters are optional.

vector: a quantity with size and direction

component: the part of a vector along one chosen axis

resultant: one vector equal to the sum of other vectors

01 Retrieve and distinguish

A cart moves from **0 m** to **+6 m** in **3 s**, then to **+2 m** in **2 s**.
1. **Find average speed** and average velocity for the full trip.
2. **Sketch the route** and **explain why the answers differ**.

Word help: velocity = rate of position change, with direction

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

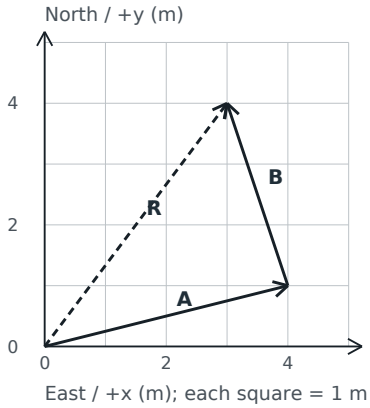
02 Explain a worked example

Reference: $R = \sqrt{(R_x^2 + R_y^2)}$.

Worked reasoning: *East*=+x; *north*=+y. R_x=**4**-**1**=**3 m**; R_y=**1**+**3**=**4 m**; R=√(**3²+4²**)=**5 m**.
θ=tan⁻¹(**4/3**)=**53.1°** *north* of *east*.

Your turn: 1. **Find B’s *westward* component**.
2. **Explain why R begins** at the original start.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Marzano check - circle: ① ② ③

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Continue down each column.

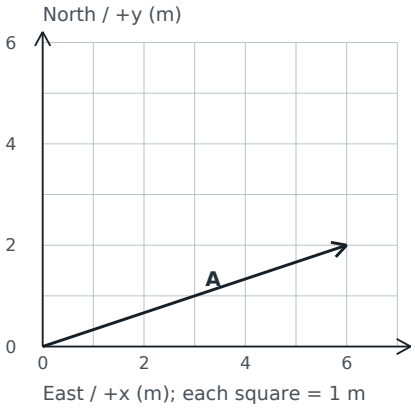
Sentence starters are optional. You may plan in any language.

03 Complete, then check

A robot makes displacement A = (**6**, **2**) m, then B = (**-4**, **3**) m. *East* is +x; *north* is +y. A is drawn.
1. **Add B tip-to-tail** and **draw the resultant** from the original start.
2. **Find its components**, magnitude and direction.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



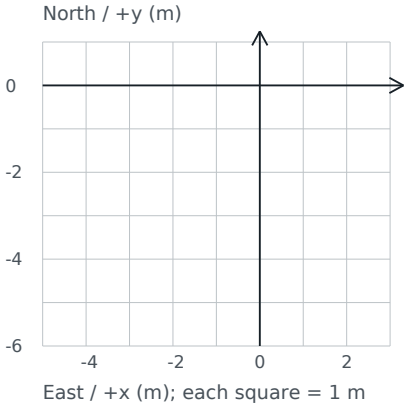
Marzano check - circle: ① ② ③

04 A different quadrant

A rover makes two moves: A = (**2**, **-5**) m and B = (**-6**, **2**) m. *East* is +x; *north* is +y.
1. **Predict the endpoint quadrant**, then **draw the route**.
2. **Calculate the resultant components**, magnitude and direction; **check them against the sketch**.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I predict … . I used … to make this prediction because … .



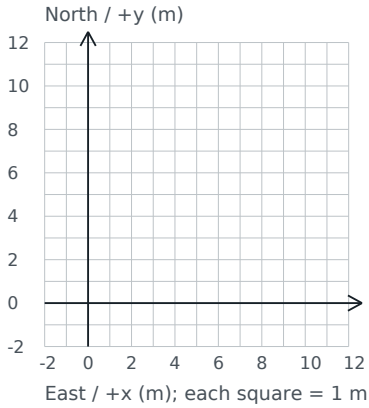
Marzano check - circle: ① ② ③

05 Choose the correct component signs

A is **10 m** at **30°** *north* of *east*: A_x = **10** cos **30°** = **8.66 m** and A_y = **10** sin **30°** = **5.00 m**.
B is **6 m** at **60°** *north* of *west*. **Sketch both arrows**, resolve B into signed components, and **find the resultant magnitude** and direction.

Word help: resolve = separate a vector into components

Sentence starter: I found … [unit]. This value means … in this situation.



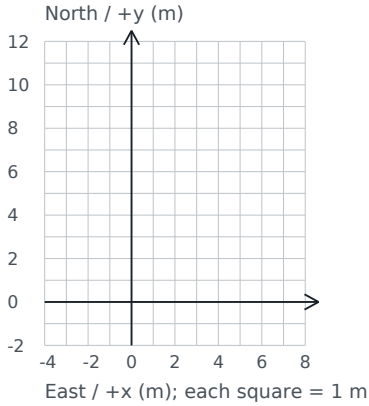
Marzano check - circle: ① ② ③

06 The angle is measured from north

Walk **12 m** at **25°** *east* of *north*, then **7 m** *west*.
1. **Sketch first** and **predict the resultant quadrant**.
2. **Calculate components**, magnitude and direction.
3. **Explain which component uses cosine** and why.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I predict … . I used … to make this prediction because … .



Marzano check - circle: ① ② ③

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Continue down each column.

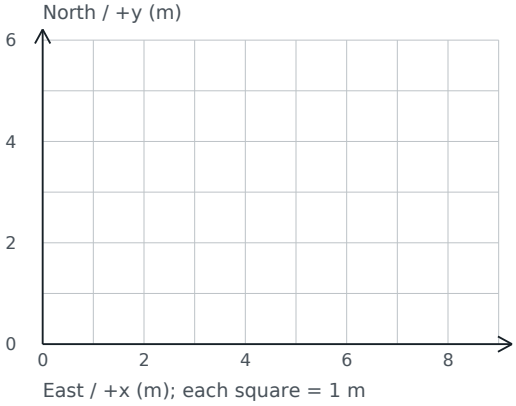
Sentence starters are optional. You may plan in any language.

07 Plan the direct return

A survey team walks **8 m** *east*, **5 m** *north*, then **2 m** *west*. A clear field allows a straight return to the start.
1. **Find the route distance before** returning and the direct return displacement, including magnitude and direction.
2. **Draw both the outward route** and the return.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.

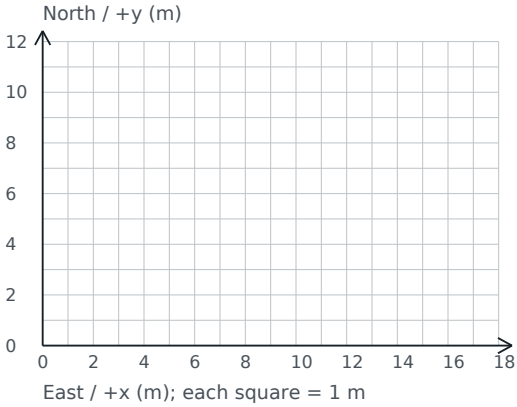


Marzano check - circle: ① ② ③

08 Diagnose the reasoning

A student adds **10 m** *east* and **10 m** at **60°** *north* of *east* using √(**10²** + **10²**) = **14.1 m**.
1. **Explain why this method** does not apply to these vectors.
2. **Draw a correct vector diagram** and use it to repair the solution. Include direction.

Sentence starter: The step I would change is … . I would change it because I found … .



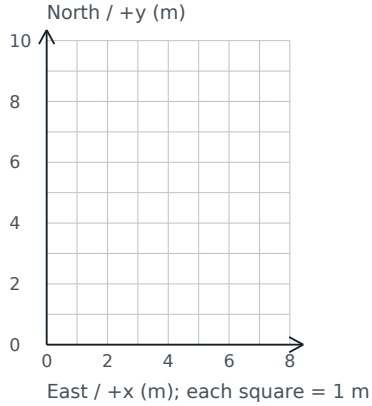
Marzano check - circle: ① ② ③

09 Represent one trip in several ways

A robot moves steadily from (**0**, **0**) m to (**6**, **0**) m in **3 s**, then steadily to (**6**, **8**) m in another **4 s**.
(a) **Draw the path** and resultant; **find distance** and displacement.
(b) **Find average speed** and the average-velocity vector for the full trip.
(c) **Sketch x-versus-time** and y-versus-time graphs. **Explain how each graph records** the same motion as your path diagram.

Word help: displacement = change from initial position to final position, with direction

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



Marzano check - circle: ① ② ③

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Continue down each column.

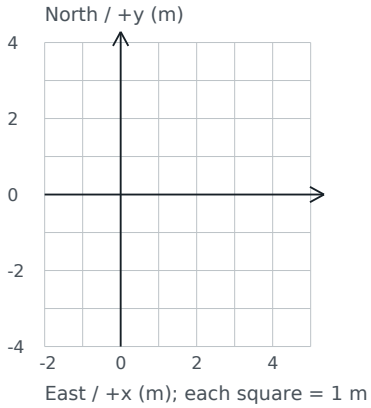
Sentence starters are optional. You may plan in any language.

10 Exit check - solve without the worked example

A = (**4**, **-3**) m and B = (**-1**, **5**) m.
1. **Find R**, its magnitude and direction.
2. **Choose C** so A + B + C = **0**.
3. **Draw a closed tip-to-tail route** and **give one physical or mathematical** check of your answer.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



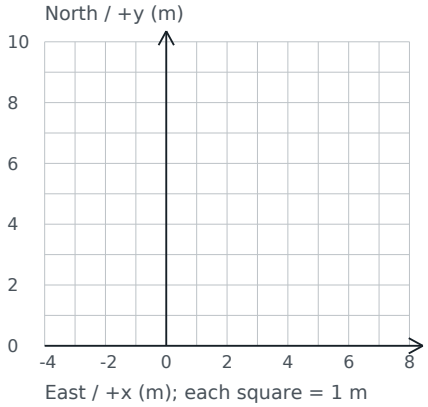
Marzano check - circle: ① ② ③

11 STRETCH: Missing direction

The first displacement is **6 m** *east*. The second has magnitude **10 m**. Finish directly *north* of the start.
1. **Find the second vector's components** and direction, and the resultant.
2. **Draw the vector diagram**.
3. **Explain whether** more than one solution meets the *north* requirement.

Word help: magnitude = size of a quantity without its direction

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .



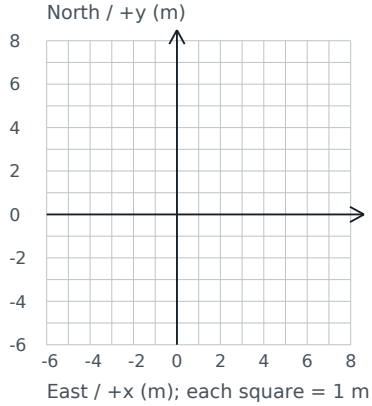
Marzano check - circle: ① ② ③

12 STRETCH: Decide what is possible

Two displacements have magnitudes **5 m** and **3 m**. **Test whether** resultants of **1 m**, **6 m** and **9 m** are possible using limits. **Draw the 6 m** case **to scale**. **Estimate the angle** between the original arrows with tails together. **Explain why the tip-to-tail triangle** uses a different interior angle. Optional: derive an angle-resultant relationship to **check your drawing**.

Word help: derive = show the reasoning or algebra that produces a relationship

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: ① ② ③